Comparative Study of Java, C, C++, Python, and JavaScript

Parameter	Java	C	C++	Python	JavaScript
Open Source or Commercial	Open-source implementation	Open source	Open source	Open source	Open source
Compiler or Interpreter	Compiled to bytecode + JIT	Compiled	Compiled	Interpreted/Bytecode	Interpreted/JIT
OOP Support	Yes	No	Yes	Yes	Yes
Developer Organization	Oracle	ANSI/ISO	ISO	Python Software Foundation	ECMA International
Developer (Person)	James Gosling	Dennis Ritchie	Bjarne Stroustrup	Guido van Rossum	Brendan Eich
Current Major Version (2026)	Java 25	C23	C++23	Python 3.14	ECMAScript 2026
Primary Purpose	General-purpose	System programming	High-performance software	General-purpose	Web scripting
Common Applications	Enterprise apps	OS, drivers	Games, systems	AI, automation	Web apps
Database Support	Excellent via JDBC	Libraries required	Libraries required	Extensive libraries	Strong via APIs
Memory Management	Garbage Collection	Manual	Manual/Smart pointers	Garbage Collection	Garbage Collection
Security Features	Sandbox, JVM security	Limited	Moderate	High-level protections	Browser sandbox
Performance	High	Very High	Very High	Moderate	High
Platform Independence	High (JVM)	Low	Medium	High	High
Other Important Features	Rich ecosystem	Low-level control	STL, templates	Simple syntax	Runs in browsers
Ease of Learning	Moderate	Moderate	Difficult	Easy	Easy
Popular IDEs	IntelliJ, Eclipse	VS Code, Code::Blocks	Visual Studio, CLion	PyCharm, VS Code	VS Code, WebStorm
Compilation Output	Bytecode (.class)	Machine code	Machine code	Bytecode (.pyc)	Engine-executed code
Advantages	Portable, scalable	Fast, efficient	Powerful, fast	Readable, productive	Web ubiquity
Limitations	Higher memory use	No OOP	Complex syntax	Slower execution	Browser differences

Unique Features:

Java: JVM portability and enterprise ecosystem.

C: Direct hardware access and efficiency.

C++: Templates, STL, and high performance.

Python: Simplicity and AI/ML ecosystem.

JavaScript: Native browser execution and full-stack web development.

Most Suitable Languages by Domain

- System Programming: C
- Enterprise Applications: Java
- Web Development: JavaScript
- Artificial Intelligence: Python
- Mobile Application Development: Java (Android ecosystem)

Conclusion

Each programming language serves different goals. C remains the preferred choice for system programming because of its speed and direct hardware access. C++ extends C with object-oriented and generic programming features, making it suitable for high-performance applications such as games and simulations. Java continues to be a leading option for enterprise software due to its portability, scalability, and strong ecosystem. Python stands out in artificial intelligence, data science, and automation because of its simple syntax and extensive libraries. JavaScript dominates web development by enabling interactive applications in browsers and on servers. The best language depends on the project requirements, performance needs, and development goals.